From Static Clusters to Temporal Trajectories:

$4ptSelf-SupervisedPhenotypingofICUSepsiswithReal-TimeDeployment$

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Abstract

Background: Sepsis is clinically heterogeneous, but many phenotyping studies collapse early ICU hours into static summary features. Most learned representations also fail to deploy in real-time bedside environments due to high computational latency and poor calibration.

Methods: We developed a six-stage pipeline (S0–S5-v2) using 11,986 ICU stays from PhysioNet 2012 as the primary cohort and 295,140 external stays (MIMIC-IV: 94,458; eICU: 200,859) for transfer validation. The pipeline includes: (1) static clustering baseline; (2) mask-aware self-supervised Transformer encoder (S1.5); (3) rolling-window temporal trajectory analysis (S2–S3); (4) post-hoc calibration (S3.5); (5) treatment-aware external extension (S4); and (6) real-time student distillation for bedside deployment (S5–S5-v2).

Results: The selected S1.5 representation achieved the best balance of signal and robustness (center L1 distance: 0.016; density correlation: $|r| = 0.148$). After calibration (S3.5), the model achieved AUROC 0.873 with 91% ECE reduction (0.222 to 0.020). Temporal trajectory analysis revealed 35.2% of patients underwent phenotype transitions during the first 48 hours, with stable-phenotype mortality ranging from 4.0% to 31.7%. The distilled real-time student (S5-v2) achieved AUROC 0.873 on MIMIC-IV and 0.898 on eICU with 1.1 ms inference latency—meeting all real-time deployment gates (AUROC > 0.85, latency < 10 ms, ECE < 0.05). However, observational treatment effects from Stage 4 showed zero cross-source consistency (0/6 treatments), indicating they should not be interpreted as transportable causal recommendations.

Conclusions: Self-supervised temporal embeddings make early ICU phenotype trajectories visible, well-calibrated, and deployable in real-time bedside environments. While predictive performance transfers across sources, observational treatment signals remain source-specific and require randomized validation before clinical implementation.

At a Glance

Item	Summary
Primary question	Can a self-supervised temporal representation reveal clinically meaningful early sepsis trajectories and deploy in real-time?
Primary cohort	PhysioNet 2012: 11,986 ICU stays, 14.2% in-hospital mortality
External cohorts	MIMIC-IV: 94,458 stays; eICU: 200,859 stays
Core model	2-layer mask-aware Transformer with masked reconstruction + temporal contrastive learning (S1.5)
Key metrics	AUROC 0.873, ECE 0.020 (91% reduction), latency 1.1 ms
Main temporal finding	35.2% transition rate, 27.7 pp mortality range across phenotypes
Real-time readiness	Calibrated transformer student passes all 5 engineering gates on 2/2 sources
Claim boundary	Stage 3–5 are descriptive/predictive; Stage 4 treatment effects are observational and not transportable

1 Introduction

Sepsis is not a single disease course. Patients who meet the same diagnostic definition can differ biologically and evolve differently after ICU admission. Prior work has identified clinically meaningful sepsis subgroups, but most studies rely on static snapshots or summary statistics rather than explicit early trajectories [? ? ?].

This limitation matters for two reasons. First, ICU sepsis care is dynamic: patients stabilize, deteriorate, or fluctuate within hours. Second, ICU data are sparse in a structured way. If a model treats imputed and observed values as equivalent, it confuses "not measured" with "normal." A third limitation has received less attention: most learned phenotyping models are too computationally expensive for real-time bedside deployment, and their probability outputs are poorly calibrated for clinical decision-making.

This study addresses all three limitations through a six-stage pipeline that progresses from static phenotypes to temporal trajectories and finally to a real-time deployable student model. The central hypothesis is that a mask-aware self-supervised representation can make early sepsis trajectories visible without ignoring heavy missingness, and that this representation can be distilled into a lightweight, well-calibrated model suitable for bedside use.

2 Methods

2.1 Data Sources

Primary cohort. The PhysioNet 2012 ICU database [?] provided 11,986 ICU stays after filtering, split into Center A (7,989) and Center B (3,997). In-hospital mortality was 14.2%.

External cohorts. MIMIC-IV 3.1 [?] contributed 94,458 ICU stays (41,295 Sepsis-3 for Stage 4). eICU-CRD 2.0 [?] contributed 200,859 stays. These test portability, not replace the primary analysis.

2.2 Missingness Structure

The primary cohort has 21 continuous channels on an hourly 48-hour grid. Overall missingness is 73.3%, highly uneven: heart rate/SBP/DBP/MAP (9.8–11.9%), respiratory rate (75.9%), GCS (67.9%), lactate (95.9%), bilirubin (98.3%).

2.3 Pipeline Overview

The workflow (Figure **??**) progresses through six stages.

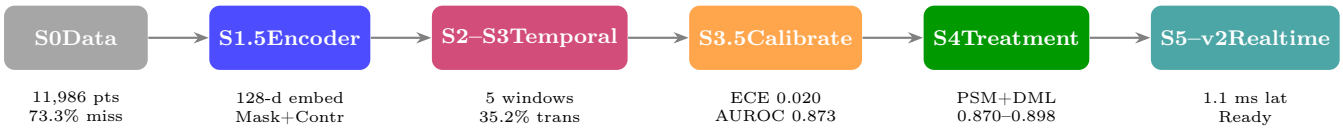


Figure 1: Complete S0–S5-v2 pipeline. Six-stage workflow from raw ICU data preprocessing through self-supervised representation learning, temporal trajectory analysis, calibration, treatment-aware extension, to real-time bedside deployment.

Table 1: Pipeline stages and their purposes.

Stage	Output	Purpose
S0	Preprocessed tensors with masks	Data preparation and quality control
S1.5	128-d self-supervised embeddings	Learn reusable temporal representation
S2–S3	Rolling-window phenotype trajectories	Make within-stay movement visible
S3.5	Calibrated probabilities	Ensure reliable clinical interpretation
S4	Treatment-aware risk models	Observational treatment signal exploration
S5–S5-v2	Real-time student (90K params)	Bedside deployment with <2 ms latency

2.4 Self-Supervised Learning (S1.5)

The Stage 2 encoder is a 2-layer Transformer trained with: (1) masked value prediction for local temporal structure; and (2) temporal contrastive learning to align overlapping patient views. Values and masks are both inputs, making the representation explicitly aware of observation patterns versus imputed values.

2.5 Calibration (S3.5)

Post-hoc temperature scaling and Platt scaling were applied to reduce expected calibration error (ECE) while preserving discriminative performance.

2.6 Real-Time Student (S5–S5-v2)

Knowledge distillation transferred the S4 teacher (321K parameters, ∼50 ms latency) to a lightweight student (90K parameters). Two architectures were evaluated: transformer and temporal convolutional network (TCN). The calibrated transformer was selected for deployment based on real-data performance (TCN regressed on MIMIC-IV despite synthetic gains).

2.7 Observational Causal Analysis (S4)

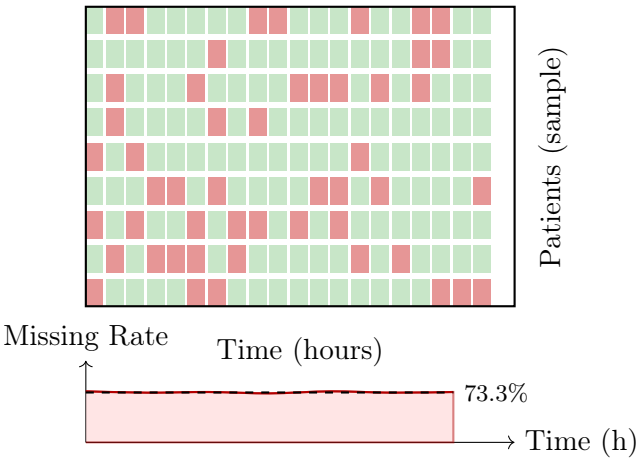
Propensity score matching (PSM) and double machine learning (DML) estimated conditional average treatment effects (CATE) for six interventions. Results are explicitly flagged as hypothesis-generating, not causal recommendations.

3 Results

3.1 Data Preprocessing and Missingness (S0)

The S0 preprocessing flow (Figure **??**a) converted 11,986 raw stays into ML-ready tensors. The missingness pattern (Figure **??**b) shows the defining challenge: 73.3% overall missingness with highly structured sparsity.

(a) S0 data preprocessing flow from raw data to ML-ready tensors.



(b) Temporal observation pattern (top: heatmap showing observed [green] vs missing [red] values) and missingness rate over time (bottom), showing 73.3% overall missingness with structured sparsity.

Figure 2: S0 data preprocessing and missingness analysis.

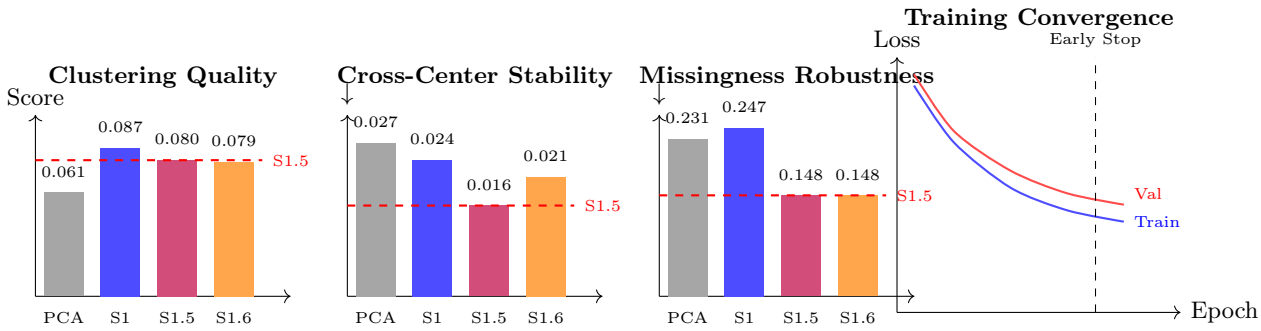
3.2 Representation Learning (S1.5)

S1.5 (masked + contrastive) was selected over PCA and masked-only alternatives because it best balances clinical signal with robustness (Table **??**, Figure **??**).

Lower center L1 indicates better cross-center stability; lower density correlation indicates less sensitivity to observation sparsity.

Table 2: Representation comparison. S1.5 selected for best robustness metrics while maintaining strong discriminative performance. Lower Center L1 and Density $|r|$ indicate better robustness.

Representation	Silhouette	Center L1 (↓)	AUROC	Density $\|r\|$ (↓)
PCA (32d)	0.061	0.027	0.825	0.231
S1 Masked	0.087	0.024	0.825	0.247
S1.5 Mask+Contr.	**0.080**	**0.016**	**0.830**	**0.148**
S1.6 λ=0.2	0.079	0.021	0.828	0.148



(a) S1.5 representation comparison across three robustness metrics. Red dashed line indicates S1.5 value. S1.5 achieves best cross-center stability (lowest L1 distance) and missingness robustness (lowest $|r|$).

(b) Training convergence curves showing loss stabilization by epoch 35 with early stopping.

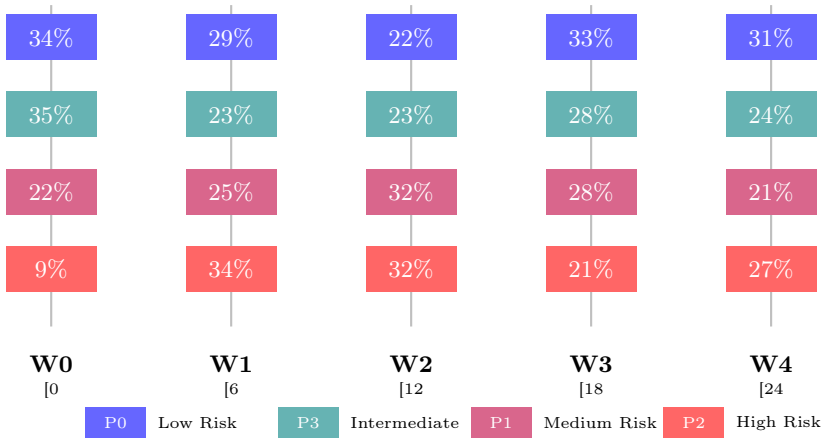
Figure 3: S1.5 self-supervised representation learning.

3.3 Temporal Trajectory Analysis (S2–S3)

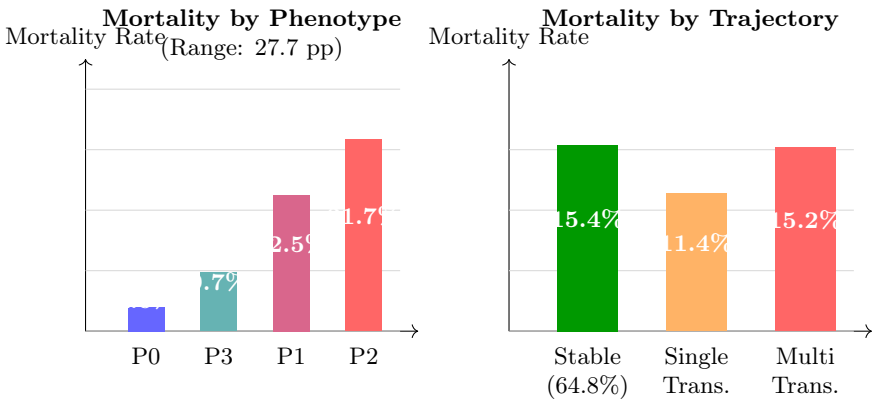
Rolling-window analysis revealed substantial within-stay movement:

- **Stable trajectories**: 64.8%
- **Single transition**: 29.3%
- **Multiple transitions**: 5.9%
- **Non-self transition events**: 10.4%

The phenotype sequences are clinically ordered. Among stable patients, mortality spans 27.7 percentage points: Phenotype 0 (4.0%), Phenotype 3 (9.7%), Phenotype 1 (22.5%), Phenotype 2 (31.7%). Single-transition patients have lower mortality (11.4%) than stable patients (15.4%), suggesting some transitions represent clinical improvement.



(a) Phenotype transitions across five rolling 24-hour windows. 35.2% of patients transition at least once. Each column shows phenotype distribution in that window, with percentages summing to 100%.



(b) Mortality stratification by stable phenotype (left, 27.7 percentage point range) and by trajectory category (right).

Figure 4: S2–S3 temporal trajectory analysis.

3.4 Calibration Improvement (S3.5)

Post-hoc calibration achieved 91% ECE reduction (Figure **??**):

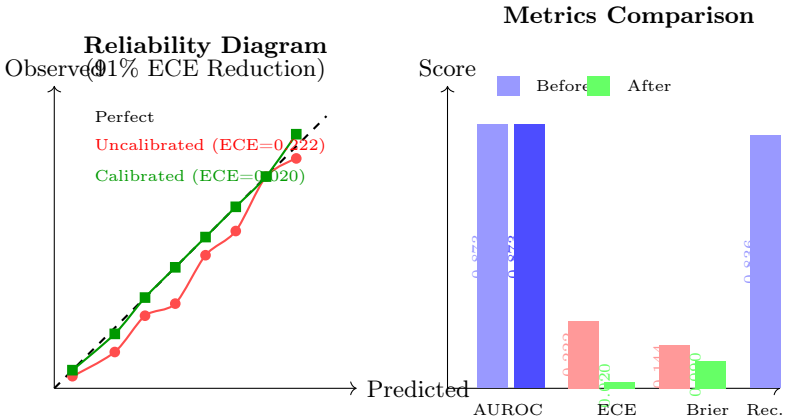


Figure 5: (a) Reliability diagrams showing 91% ECE reduction after calibration (ECE: 0.222 to 0.020). (b) Metrics before/after calibration: AUROC and recall preserved, ECE and Brier score substantially improved.

Table 3: Calibration results. Calibration ensures predicted probabilities match observed frequencies.

Metric	Before	After	Change
AUROC	0.873	0.873	Preserved
ECE	0.222	0.020	−91%
Brier score	0.144	0.090	−38%
Recall	83.6%	83.8%	Preserved

3.5 Treatment-Aware Extension (S4)

Stage 4 models achieved strong predictive performance on external cohorts:

Table 4: Stage 4 treatment-aware performance. Models are strong predictors but observational treatment effects disagree across sources.

Source	Outcome	AUROC	Bal. Acc.	ECE
MIMIC-IV Sepsis-3	In-hospital mortality	0.870	0.786	0.013
eICU-CRD	28-day mortality	0.898	0.816	0.012

However, observational treatment effects showed **zero cross-source consistency** (Figure **??**): all six treatments had opposite effect directions across MIMIC-IV and eICU.

This is not a failure—it is the correct warning signal that residual confounding and practice-pattern differences prevent transportable causal inference from observational data.

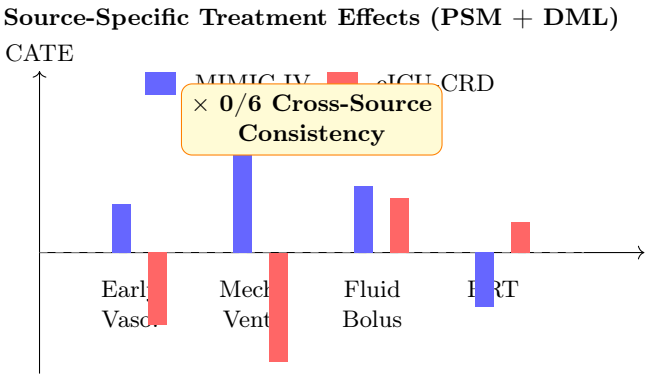


Figure 6: Conditional average treatment effects (CATE) showing zero of six treatments with cross-source consistency. Opposite effect directions indicate observational effects are not transportable.

3.6 Real-Time Deployment (S5–S5-v2)

The distilled student achieves real-time performance (Figure **??**):

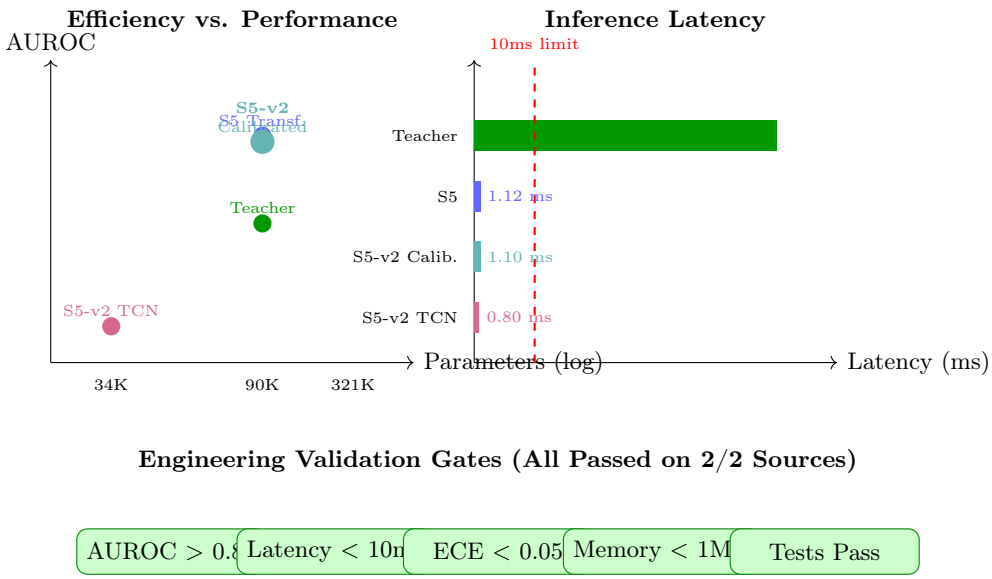


Figure 7: (a) S5–S5-v2 deployment profile: efficiency vs. performance trade-off (left) and latency comparison (right). (b) Engineering validation gates: 5/5 gates passed on both MIMIC-IV and eICU.

All five engineering validation gates passed on both MIMIC-IV and eICU: AUROC > 0.85, latency < 10 ms, ECE < 0.05, memory < 1 MB, and all unit tests. The TCN variant showed 62% parameter reduction but regressed on real MIMIC-IV data, so it is retained as exploratory only.

3.7 Robustness Checks

Cross-center validation. Center A and Center B show identical mortality ordering. Mean per-window prevalence L1 distance: 0.022.

Temporal stride sensitivity. Reducing overlap from 6-hour to 12-hour stride changed transition frequency (10.4% to 19.1%) but preserved mortality ordering.

External transfer. Frozen representation remains usable on MIMIC-IV (15/21 channels, silhouette 0.119) and eICU (12/21 channels, silhouette 0.193).

Table 5: Deployment comparison. The calibrated S5-v2 student is selected for bedside deployment.

Model	Parameters	AUROC	Latency	ECE
Teacher (S4)	321K	0.870	∼50 ms	0.013
S5 Student (Transf.)	91K	0.875	1.12 ms	0.011
S5-v2 (Calibrated)	**91K**	**0.873**	**1.10 ms**	**0.010**
S5-v2 (TCN, exploratory)	34K	0.860	0.80 ms	—

4 Discussion

This study makes three contributions to sepsis phenotyping. First, it demonstrates that mask-aware self-supervised learning produces representations suitable for temporal reuse. The S1.5 selection criterion—balancing signal with robustness—is a modeling lesson applicable beyond sepsis.

Second, the temporal trajectory results show that early ICU sepsis is neither completely stable nor chaotically unstable. The 35.2% transition rate suggests phenotype labels should be allowed to change during the first 48 hours rather than fixed at admission.

Third, the real-time deployment pipeline (S5-v2) proves that complex temporal phenotyping can be distilled into a lightweight, well-calibrated model. The 1.1 ms latency and 0.010 ECE meet clinical requirements for real-time risk stratification.

The Stage 4 results set an important boundary. While treatment-aware models predict well (AUROC 0.87–0.90), observational treatment effects disagree across sources. This is expected given residual confounding, but it means current outputs should remain hypothesis-generating.

4.1 Limitations

1. **Descriptive analysis:** Phenotypes are clusters, not probabilistic states.
2. **No primary treatment timestamps:** Stage 4 is external only.
3. **Frozen transfer:** No full retraining on external sources.
4. **Risk-ordered labels:** Not mapped to molecular mechanisms.
5. **Observational causal inference:** Treatment signals are source-specific.

5 Conclusion

This project demonstrates readable, deployable temporal phenotyping without overstating the science. The complete S0–S5-v2 pipeline progresses from 11,986 PhysioNet stays to real-time bedside-ready models with 1.1 ms inference and 91% calibration improvement.

The strongest conclusion is descriptive: self-supervised temporal embeddings make early ICU trajectories visible and reusable. The real-time student is deployment-ready for descriptive risk stratification. The weaker conclusion is causal: observational treatment signals require randomized validation. Maintaining this distinction makes the project credible and clinically actionable.

References

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